

Employee Management System SQL Exercises

Database Schema

The following schema defines the structure for an Employee Management System:

Departments Table

```
CREATE TABLE Departments
( DepartmentID INT PRIMARY KEY,
  DepartmentName VARCHAR(100)
);
```

Employees Table

```
CREATE TABLE Employees
( EmployeeID INT PRIMARY KEY,
  FirstName VARCHAR(50),
  LastName VARCHAR(50),
  DepartmentID INT FOREIGN KEY REFERENCES Departments(DepartmentID),
  Salary DECIMAL(10,2),
  JoinDate DATE
);
```

Sample Data

The following sample data can be used for testing:

Departments Sample Data

```
INSERT INTO Departments (DepartmentID, DepartmentName) VALUES
(1, 'HR'),
(2, 'Finance'),
(3, 'IT'),
(4, 'Marketing');
```

Employees Sample Data

```
INSERT INTO Employees (EmployeeID, FirstName, LastName, DepartmentID, Salary,
JoinDate) VALUES
(1, 'John', 'Doe', 1, 5000.00, '2020-01-15'),
(2, 'Jane', 'Smith', 2, 6000.00, '2019-03-22'),
```

```
(3, 'Michael', 'Johnson', 3, 7000.00, '2018-07-30'),  
(4, 'Emily', 'Davis', 4, 5500.00, '2021-11-05');
```

Exercises

Exercise 1: Create a Stored Procedure

Goal: Create a stored procedure to retrieve employee details by department.

Steps:

1. Define the stored procedure with a parameter for DepartmentID.
2. Write the SQL query to select employee details based on the DepartmentID.
3. Create a stored procedure named `sp\_InsertEmployee` with the following code:

Solution:

```
CREATE PROCEDURE sp_GetEmployeesByDepartment  
    @DepartmentID INT  
AS  
BEGIN  
    SELECT  
        EmployeeID,  
        FirstName,  
        LastName,  
        DepartmentID,  
        Salary,  
        JoinDate  
    FROM Employees  
    WHERE DepartmentID = @DepartmentID;  
END;
```

```
EXEC sp_GetEmployeesByDepartment 1;
```

Exercise 4: Execute a Stored Procedure

Goal: Execute the stored procedure to retrieve employee details for a specific department.

Steps:

1. Write the SQL command to execute the stored procedure with a DepartmentID parameter.
2. Execute the command and review the results.

Solution:

```
EXEC sp_GetEmployeesByDepartment  
    @DepartmentID = 2;
```

Exercise 5: Return Data from a Stored Procedure

Goal: Create a stored procedure that returns the total number of employees in a department.

Steps:

1. Define the stored procedure with a parameter for DepartmentID.
2. Write the SQL query to count the number of employees in the specified department.
3. Save the stored procedure by executing the Stored procedure content.

Solution:

```
CREATE PROCEDURE sp_GetEmployeeCountByDepartment
```

```
    @DepartmentID INT
```

```
AS
```

```
BEGIN
```

```
    SELECT
```

```
        COUNT(*) AS TotalEmployees
```

```
    FROM Employees
```

```
    WHERE DepartmentID = @DepartmentID;
```

```
END;
```

```
EXEC sp_GetEmployeeCountByDepartment
```

```
    @DepartmentID = 3;
```